DATA ITEM DESCRIPTION

Title: NUCLEAR SURETY EVALUATION (NSE)

Number: DI-NUOR-82401
AMSC Number: F10376
DTIC Applicable: No
Preparing Activity: 37 (AFSEC/SEWN)
Applicable Forms: N/A

Approval Date: 20230203
Limitation: N/A
GIDEP Applicable: No
Project Number: NUOR-2022-001

Use/Relationship: A Nuclear Surety Evaluation (NSE) is a technical report developed by a nuclear weapons System Program Office (SPO) to document compliance to all Nuclear Safety Design Certification (NSDC) requirements identified in a respective Certification Requirements Plan (CRP). The CRP defines the requirements, assigns the roles and responsibilities, and defines all the activities and tasks required to achieve Nuclear Certification. The CRP is created to meet the needs of a particular acquisition or modification effort. As program changes are encountered and incorporated into the acquisition or modification effort, their impact on the nuclear certification process is reviewed and the CRP is updated accordingly. The Program Manager-approved CRP contains all required details regarding how and when the program office will meet the certification requirements. Nuclear Safety Design Certification requirements are listed in a Certification Requirements Matrix (CRM) within the Safety Design section of the CRP. The NSE provides the Air Force Safety Center with the analyses of the work the contractor completes to meet or exceed compliance with Nuclear Safety Design Certification requirements in Air Force Manual (AFMAN) 91-118, *Safety Design and Evaluation Criteria for Nuclear Weapon Systems* and Department of the Air Force Manual (DAFMAN) 91-119, *Safety Design Criteria for Nuclear Weapon Systems Software*, and guidance in Department of the Air Force Instruction (DAFI) 91-101, *Air Force Nuclear Weapons Surety Program*. The NSE also describes the weapon system, the specific components, assembly, or systems analyzed, methods used, and concerns resulting from analyses performed.

This Data Item Description (DID) contains the format and content preparation information for the deliverable resulting from the work task described in the solicitation and contract. This DID is related to *NPFC - DI-NUOR-81409*, *Certification Requirements Plan (CRP)*, AFMAN 91-118, DAFMAN 91-119, and DAFI 91-101. (Copies of DAFIs, DAFMANs, and AFMANs are available online at http://www.e-publishing.af.mil/.) The U.S. Navy is excluded from requirements set forth in this DID.

Requirements:

1. **Reference Documents**: The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. **Format**: The NSE shall be in contractor's format, following the outline provided within this document.

3. **Content**:

   3.1. The NSE shall be limited in scope to a single certification action but shall not limited in length. A certification action is a piece of new equipment or an already-certified piece of equipment in need

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of modification that interfaces with nuclear weapon system and requires Nuclear Safety Design Certification.

3.2. The NSE shall define the component, assembly, or system analyzed. It shall include identifying information such as item identification numbers and item uses/description.

3.3. The NSE shall include a summary of engineering analysis, demonstration, inspection, and test through technical qualitative reports, technical quantitative reports, photos, videos, diagrams, schematics, graphs, and sketches.

3.4. The NSE shall include approved deviations as outlined in DAFI 91-101. Note: all deviations must be approved prior to Nuclear Safety Design Certification being granted.

3.5. The NSE shall be specific enough to demonstrate to the Air Force Safety Center, Weapons Division (AFSEC/SEW) that the SPO meets the Nuclear Safety Design criteria agreed to in the CRP.

3.6. Outline: For hardware and/or software (as applicable) provide:

3.6.1. Certification Action Information

3.6.1.1. SPO recommendation of "Certification" or "Certification with Restrictions"

3.6.2. Item Identification

3.6.2.1. Hardware or Software designation

3.6.2.2. Nomenclature or Common Name

3.6.2.3. National Item Identification Number (NIIN)

3.6.2.4. Manufacturer and Manufacturer Code

3.6.2.5. Model/Part Number/Software number

3.6.2.6. Weapons used with

3.6.2.7. Item Manager/Nuclear Certification Manager (NCM)/POC Information

3.6.3. Uses and Description

3.6.3.1. Equipment uses

3.6.3.2. Interface with Critical Functions

3.6.3.3. Technical Orders with identified Nuclear Surety Procedures (NSP)

3.6.3.4. Relevant Non-Analytical Information

3.6.4. Summary of AFMAN 91-118 and DAFMAN 91-119 Requirement Compliance

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3.6.4.1. Requirement compliance shown through engineering analysis, demonstration, inspection, and test. Requirement compliance shall be provided in either report format or CRP Matrix format. Either format is acceptable provided the submitter addresses every CRP line item for compliance. See Appendix A of this document.

3.6.4.2. When required by the CRP, the submitter provides a Critical Signal Analysis (CSA) for DAFMAN 91-119 compliance. See Appendix B of this document.

3.6.5. Deviations and Recommended Restrictions

3.6.5.1. Provide approved Nuclear Surety Deviations for this certification action.

3.6.5.2. Provide any recommended restrictions from the summary of requirement compliance captured in the document.

3.6.6. Appendix A – Proof of Nuclear Safety Design Certification Compliance

3.6.6.1. Either in report format or CRP matrix format, identify the specific AFMAN 91-118 and/or DAFMAN 91-119 criteria. The content of the appendix shall reference line items in the compliance matrix within the CRP for traceability. The purpose is to show how the program has complied with every compliance matrix requirement specified within the CRP for AFMAN 91-118 and DAFMAN 91-119.

3.6.6.1.1. Cite each document and paragraph used (e.g., test plans, reports, bent pin analysis) to show compliance with each AFMAN 91-118 and/or DAFMAN 91-119 requirement. Discuss the verification methods used to justify the compliance.

3.6.6.1.2. These references can be submitted as attachments to the NSE.

3.6.6.1.3. Specify deficiencies against compliance for which a deviation has not been approved.

3.6.7. Appendix B – Critical Signal Analysis

3.6.7.1. Provide the following in the Critical Signal Analysis:

3.6.7.1.1. Flow of Critical Signals from human intent to critical function activation

3.6.7.1.2. All components between human intent and critical function activation

3.6.7.1.3. The potential effect those components could have on the Critical Signal

3.6.7.1.4. The protection mechanisms provided to Critical Signals

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Figure 1: NSE Template

Legend
Black Text = permanent text
Red Text = template example text

Nuclear Surety Evaluation
INSERT CERTIFICATION ACTION TITLE

This NSE was developed in accordance with the Certification Requirements Plan for Certification Action , Initial Approval: Date; Revision 1: Date… and DAFI 91-101, *Nuclear Weapons Surety Program,* Date of publication.

1. **Certification Action Information**
   1.1 Recommend: "Certification"/"Certification with Restrictions" of Certification Action
2. **Item Identification**
   2.1 Hardware/Software: HW/SW
   2.2 Nomenclature/Common Name: LRSO Missile Lift Beam / MMIII Fast-Rising B Plug / WGF X Lightning Protection System
   2.3 National Item Identification Number (NIIN): 01-234-5678
   2.4 Manufacturer and Manufacturer Code: e.g. Northrop Grumman Corporation, 96-735-6127;The Boeing Company, 12-636-8257
   2.5 Model/Part Number/Software Number: 0123-45
   2.6 Weapons: W80-4, B61-12, etc.
   2.7 Item Manager/Nuclear Certification Manager (NCM)/POC Information: (505) 123-4567, email.address@website.com
3. **Uses and Descriptions**
   3.1 This hardware/software is a tool used to… explanation of use.
   3.2 This hardware/software interfaces with the Authorization/Prearm/Release/Arm/Launch critical function(s).
   3.3 This hardware/software has Nuclear Surety Procedures within the following Technical Orders: Technical Orders.
   3.4 Relevant Non-Analytical Information: Information the SPO think might aid AFSEC in adjudicating the NSE.
4. **Summary of AFMAN 9-118 and DAFMAN 91-119 Requirement Compliance**
   4.1 Requirement compliance is shown through engineering analysis, demonstration, inspection, and test. Requirement compliance can be provided in either report format or CRP Matrix format. Either format is acceptable provided the submitter addresses every CRP line item for compliance. See Appendix A of this document.
   4.2 When required by the CRP, the submitter provides a Critical Signal Analysis (CSA) for DAFMAN 91-119 compliance. See Appendix B of this document.
5. **Deviations and Recommended Restrictions**
   5.1 The SPO is tracking the following approved deviations: Approved Deviations.
   5.2 The SPO recommends the following restrictions:
      5.2.1 Restrictions

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Figure 2: NSE Template (Continued)

Appendix A: Proof of Nuclear Safety Design Certification Compliance

1. CRP Item #1: *AFMAN 91-118, Para X.X.X: AFMAN verbiage.*
   1.1 Supporting analytical documentation (e.g., The Tooling Analysis Report, para 3.4; ANSI Standard X, para 2.3.2) provides the analysis/demonstration/inspection/test that the SPO uses to justify compliance with this safety design requirement.
   1.2 Supporting analytical documentation is included in this NSE as Appendix X. Detail the information presented in 1.1.
   1.3 The SPO complies with CRP item #1 and identifies no deficiencies /does not comply with CRP item 1 and identifies X deficiencies during analysis.
2. CRP Item #2: …
3. CRP Item #3: …

OR

PARAGRAPH/DESCRIPTION	APPLIES	METHODOLOGY	Compliance	Proof of Compliance
AFMAN 91-11X, Chapter X – Chapter Name			***NSE Analysis***	
X.X.X. Requirements text. This is where the respective AFMAN requirement is placed, verbatim.	**YES**	This methodology section is where the program labels their intended method of showing compliance with this requirement in the NSE.	**Yes, No, TBD**	*NSE Appendix B, Test Report X.X.X.* *Attachment 1, Data X.X.X* *Attachment 2,, Photos 1-4* *Tell us where the data is located in the attachment list to include paragraph.*
COMMENTS: This comment section is where the program explains their purpose for their decided method of showing compliance. Any references to industry/military standards intended to supplement the compliance also goes here.			COMMENTS: This comment section is where the program provides a small snippet of information regarding the results of their analysis.	

Appendix B: Critical Signal Analysis

1. The Critical Signal Analysis will provide the following:
   1.1 The flow of Critical Signals from human intent to critical function activation.
   1.2 All components between human intent and critical function activation
   1.3 The potential effect those components could have on the Critical Signal
   1.4 The protection mechanisms provided to Critical Signals

End of DI-NUOR-82401.

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